

# An Educational Prototype of an Artificial Pancreas for Neurodivergent Learning

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## Abstract

This project presents an educational prototype of an artificial pancreas, or insulin pump, developed within the IINTS vision of accessible science education. The device is not designed or certified for insulin administration in humans. Instead, it is a safe physical and digital model that demonstrates how complex diabetes-care technologies work. Internally, the prototype uses scientifically grounded computational models, including bi-exponential pharmacokinetic simulation, hydraulic volume calculations, PID closed-loop control, and state-feedback safety logic. Its educational purpose is twofold: to show that advanced medical algorithms can run on low-cost microcontrollers, and to provide a low-sensory interface that helps neurodivergent children understand Type 1 Diabetes in a safe simulation environment.

## 1 The History and Educational Vision of IINTS

Before discussing the technical design, it is important to define the context in which this prototype was created. The project was developed under the IINTS initiative. IINTS was not founded to compete with medical-device manufacturers, nor to produce illegal or unregulated medical devices. Its purpose is education, awareness, and scientific accessibility.

The pumps developed within IINTS are educational models. They are safe physical and digital prototypes designed to help patients, classmates, teachers, and medical students understand the technology behind diabetes care. By reconstructing a life-saving but often poorly understood device, the project demystifies the science of insulin pumps, artificial pancreas systems, and closed-loop control.

This particular prototype focuses on a group that is often overlooked in diabetes education: neurodivergent children. Instead of presenting diabetes technology only through abstract numbers and clinical terminology, the model uses visual, low-sensory interaction to make invisible biological processes more understandable.

## 2 Introduction

Type 1 Diabetes (T1D) is a chronic autoimmune condition in which the pancreas produces little or no insulin [1]. As a result, patients must learn how blood glucose, carbohydrates, insulin, physical activity, and timing interact. For children, this is already difficult; for neurodivergent children, the learning process can be made harder by cognitive overload, sensory stress, and interfaces that rely heavily on abstract numerical information.

Modern insulin pumps, continuous glucose monitors, and automated insulin-delivery systems are powerful technologies in diabetes care [2], but they are often experienced as closed, clinical, and difficult to understand. This can create distance between the user and the science that affects their daily life. The educational problem addressed by this project is therefore not only technical, but also pedagogical: how can complex diabetes technology be made visible, safe, and understandable?

This project proposes a low-sensory, open-source educational artificial pancreas prototype built around the Raspberry Pi Pico microcontroller. The prototype does not deliver insulin to humans. Instead, it simulates insulin delivery, glucose trends, and safety decisions in a controlled learning environment. Its purpose is to show that an affordable embedded system can combine realistic actuator mathematics, pharmacokinetic reasoning, and neurodivergent-friendly human-computer interaction.

### 3 Hardware Architecture

The educational prototype, named IINTS MK2, integrates a small set of low-cost embedded components selected for accessibility, reproducibility, and computational sufficiency. The central controller is a Raspberry Pi Pico based on the RP2040 microcontroller [18]. This platform is compact, inexpensive, widely documented, and powerful enough to run the required medical calculations in MicroPython while also controlling the display, buttons, and motor driver.

Motor actuation is handled through a DRV8833 motor driver [19], which provides bidirectional control of the pump actuator while isolating the microcontroller from the higher current requirements of the motor. Visual feedback is provided through an ST7789 TFT LCD screen [20]. This display allows the prototype to present both numerical information and simplified visual representations for the low-sensory interface mode. Together, these components create a physically realistic embedded platform for demonstrating artificial pancreas control logic without clinical use.



**Figure 1.** Physical IINTS MK2 educational artificial pancreas prototype. The photograph shows the compact IINTS MK2 pump model used to demonstrate actuator control, display feedback, and low-sensory interaction in a safe learning context.

## 4 Software Architecture

Medical software must remain responsive at all times. A pump interface should not freeze while the device waits for a sensor update, redraws a screen, or calculates a bolus. For this reason, the prototype software is structured as a non-blocking state machine in MicroPython. Instead of using long `sleep()` delays, the program compares timestamps using `utime.ticks_ms()` and `utime.ticks_diff()`.

This architecture allows the pump to update the display, respond to buttons, process simulated CGM values, and evaluate safety formulas concurrently within the main loop. Listing 1 shows the timer-based structure used for real-time CGM simulation.

Listing 1: Non-blocking state machine for real-time CGM simulation

```
# Listing 1: Non-blocking state machine for real-time CGM simulation
def update_cgm_sensor(force=False):
    global huidige_bg, cgm_index, cgm_last_update_ms
    now = utime.ticks_ms() # Get current time in milliseconds
    elapsed_ms = utime.ticks_diff(now, cgm_last_update_ms)

    # Update only if 60,000 ms (1 minute) have passed, without blocking the CPU
    if force or elapsed_ms >= CGM_UPDATE_MS:
        steps = max(1, elapsed_ms // CGM_UPDATE_MS)
        cgm_index = (cgm_index + steps) % len(CGM_PATTERN)
        cgm_last_update_ms = now
        huidige_bg = CGM_PATTERN[cgm_index]
        return True

    return False
```

As seen in Listing 1, the simulated sensor update is event-driven rather than delay-driven. This ensures that the processor remains available for interface updates and safety checks during every cycle of the program.

## 5 Biomechanics and Fluid Dynamics of the Actuator

Although the prototype is educational and not intended for human insulin delivery, realistic actuator mathematics are essential for demonstrating how real pumps calculate micro-doses. The stepper motor therefore imitates the fluid dynamics of a syringe-based pump using principles from hydraulics and cylindrical geometry.

### 5.1 Hydraulic Transmission: Pascal's Principle

Insulin, as an aqueous solution, can be treated as practically incompressible in a first-order educational approximation. In the idealised model used here, the expelled volume is assumed to equal the mechanical reduction in reservoir volume. In real infusion systems, this relationship can be affected by compliance in the reservoir and tubing, backlash in the lead screw, trapped air, friction, pressure losses, and missed motor steps. The actuator in this project consists of a bipolar stepper motor connected to a lead screw that compresses a syringe-like reservoir.

## 5.2 Mathematical Volume and Dose Calculation

The conversion from GPIO signal to insulin-unit equivalent is modelled with the following assumptions. The dose values assume a 200-step motor; if an  $18^\circ$  per step micro linear stepper is used instead, the motor has only 20 steps per revolution and the per-step dose becomes ten times larger.

- **Stepper motor assumption:**  $N_{\text{steps}} = 200$  steps per revolution, corresponding to  $1.8^\circ$  per step.
- **Lead screw pitch:**  $p = 0.5$  mm of linear displacement per revolution.
- **Reservoir geometry:** internal diameter 9.5 mm, giving  $r = 4.75$  mm.

The volume conversion is:

$$V_{\text{rev}} = \pi r^2 p = \pi \cdot (4.75 \text{ mm})^2 \cdot 0.5 \text{ mm} \approx 35.44 \text{ mm}^3 = 35.44 \text{ } \mu\text{L},$$

$$V_{\text{step}} = \frac{V_{\text{rev}}}{N_{\text{steps}}} = \frac{35.44 \text{ } \mu\text{L}}{200} \approx 0.177 \text{ } \mu\text{L}/\text{step}.$$

For U-100 insulin, 1 unit = 10  $\mu\text{L}$ . The insulin-unit conversion is therefore:

$$D_{\text{rev}} = \frac{35.44 \text{ } \mu\text{L}}{10 \text{ } \mu\text{L}/\text{unit}} = 3.544 \text{ units},$$

$$D_{\text{step}} = \frac{0.177 \text{ } \mu\text{L}/\text{step}}{10 \text{ } \mu\text{L}/\text{unit}} \approx 0.0177 \text{ units}/\text{step},$$

$$\text{Steps per unit} = \frac{1}{0.0177} \approx 56.4 \text{ steps}/\text{unit}.$$

Thus, the educational chain is GPIO signal  $\rightarrow$  motor step  $\rightarrow$  linear displacement  $\rightarrow$  reservoir volume  $\rightarrow$  insulin-unit equivalent. With a 20-step motor, the equivalent value would be approximately 0.177 units/step, ten times coarser than the 200-step assumption.

## 6 Pharmacokinetics: Bi-Exponential IOB Decay Model

A key learning objective of the model is the concept of active insulin, commonly referred to as Insulin On Board (IOB). In real diabetes care, tracking IOB is essential for preventing insulin stacking and the resulting risk of hypoglycaemia. Early insulin pumps often used simplified linear decay models, which assume that insulin activity disappears at a constant rate over time.

### 6.1 The Two-Compartment Bi-Exponential Model

In real pharmacokinetics, the absorption and clearance of subcutaneously administered insulin follow a complex two-compartment process: insulin first moves from subcutaneous tissue into the bloodstream and is then eliminated from circulation. To model this more realistically, the IINTS MK2 algorithm uses a bi-exponential approximation rather than a simplified quadratic curve. This approach is consistent with compartment-based modelling principles used in artificial pancreas research [5, 6].

The simulated active-insulin response is described as:

$$\text{IOB}(t) = \text{Dose} \cdot \frac{k_a}{k_a - k_e} \left( e^{-k_e t} - e^{-k_a t} \right), \quad (1)$$

where  $k_a$  is the absorption constant and  $k_e$  is the elimination constant. The difference between these two exponential terms produces the characteristic rapid rise and long physiological tail of rapid-acting insulin.

In the MicroPython implementation, these exponential terms are calculated with the microcontroller’s mathematical library using `math.exp`. This allows the educational prototype to show that even a low-cost microcontroller can execute advanced pharmacokinetic calculations. For safety and clarity, the model is used for simulation and education only; it is not intended to determine real patient dosing.

Listing 2: Bi-exponential active-insulin calculation

```
def calculate_iob_biexponential(dose, elapsed_min, ka, ke):
    t = elapsed_min / 60.0 # minutes to hours
    if t <= 0 or abs(ka - ke) < 0.0001:
        return 0.0

    activity = dose * (ka / (ka - ke)) * (
        math.exp(-ke * t) - math.exp(-ka * t)
    )
    return max(0.0, activity)
```

Listing 2 demonstrates how the mathematical model is translated into compact embedded code. By using exponentials directly, the IINTS MK2 can visualise the pharmacokinetic tail of insulin more faithfully than a straight-line decay model.

## 7 Control Theory: PID Controller for Closed-Loop Automation

A complete artificial pancreas does more than respond to manual bolus requests. It must continuously adjust basal insulin delivery based on the current glucose value and the simulated Continuous Glucose Monitor (CGM) trendline. In the IINTS MK2, this is demonstrated through a full PID controller, a classical feedback-control structure widely used in automation and closed-loop medical-device research [6, 8].

### 7.1 Proportional, Integral, Derivative Control

The controller calculates the error  $e(t)$  between the current blood glucose value and the target value once per simulated minute. This error is then weighted through three gains:

- **Proportional (P):** the current absolute deviation from the glucose target.
- **Integral (I):** the accumulated historical deviation, with windup restrictions to prevent excessive correction.
- **Derivative (D):** the rate of change of the glucose trendline, indicating whether glucose is rising or falling.

The resulting temporary basal adjustment is calculated as:

$$\text{TempBasal}(t) = K_p \cdot e(t) + K_i \cdot \int e(t) dt + K_d \cdot \frac{de(t)}{dt}. \quad (2)$$

This algorithm forms the educational core of closed-loop automation. It shows how a system can act before a glucose value becomes dangerously high or low, because the derivative term responds to the direction and speed of the trendline rather than only the current value. In the IINTS MK2, the PID output is limited by safety boundaries so that the demonstration remains bounded and interpretable.

## 8 Pharmacodynamics: Carbohydrate Absorption and Safety Algorithm

To calculate safe bolus behaviour in the simulation, the pump must also estimate how much glucose is still being absorbed from the digestive system. This quantity is referred to as Carbs On Board (COB). The algorithm integrates all meals recorded in the `carb_log` over a medically standardised digestion time of 3 hours, or 180 minutes, for a mixed meal. A linear absorption model is applied:

$$\text{COB}(t) = \text{Carbohydrates} \cdot \left(1 - \frac{t_{\text{elapsed}}}{180 \text{ min}}\right). \quad (3)$$

### 8.1 State Feedback and Conditional IOB Deduction

To demonstrate how artificial pancreas systems avoid overcompensation and integral windup, the software uses state feedback. The model shows how safety is built into meal-related dosing through a deductible-IOB limit:

$$\text{Deductible IOB} = \max\left(0, \text{IOB} - \frac{\text{COB}}{\text{ICR}}\right), \quad (4)$$

where ICR is the insulin-to-carbohydrate ratio. This rule teaches users that active insulin is usually subtracted from a correction, except when that insulin is still needed to cover recently eaten carbohydrates. Only insulin that is effectively unassigned is deducted from the correction bolus. The final simulated dose is calculated as:

$$\text{Total Dose} = \text{CarbBolus} + \text{CorrectionBolus} - \text{Deductible IOB}. \quad (5)$$

Listing 3 shows how this safety concept is translated into MicroPython code.

Listing 3: The Deductible IOB safety algorithm

```
# Listing 3: The Deductible IOB Safety Algorithm
# We subtract active insulin (IOB) from the bolus, BUT we protect the insulin
# that is reserved for digesting the stomach contents (COB).
# This prevents dangerous under-dosing (hyperglycemia) after a meal.
cob_insulin_needed = cob / ICR
deductible_iob = max(0.0, iob - cob_insulin_needed)

# Calculate the final safe dose
berekende_dosis = carb_bolus + correction_bolus - deductible_iob
# Ensure the pump never reverses or administers a negative dose
totale_dosis = max(0.0, berekende_dosis)
```

As seen in Listing 3, the built-in `max()` function of MicroPython establishes a hard lower limit on deducted insulin and final dose output. As an educational demonstration, this formula links abstract control theory to an understandable safety concept: insulin can only be subtracted when it is not still needed for active carbohydrates.

## 9 Scientific Edge Cases and Advanced Logic

During development of the IINTS MK2 artificial pancreas simulation, several advanced clinical edge cases were identified and translated into mathematical logic. These cases are not included to enable real patient treatment, but to demonstrate that robust diabetes technology must respond to more than a single glucose value. Meals, sensor delay, circadian rhythm, and physical activity all change the meaning of the same numerical blood-glucose reading.

### 9.1 The “Pizza” Dual-Wave Bolus: Gastric Emptying

High-fat and high-protein meals can delay gastric emptying and extend postprandial glucose absorption, which is why real pump therapy often considers extended or dual-wave bolus strategies for meals such as pizza [9, 10]. In the Tamagotchi mode, selecting the “Pizza” portion automatically triggers a `DUAL_WAVE_ACTIVE` state. Mathematically, this changes the carbohydrate absorption window from the standard 3 hours to 6 hours:

$$T_{\text{abs}} = \begin{cases} 180 \text{ min,} & \text{standard meal,} \\ 360 \text{ min,} & \text{dual-wave pizza mode.} \end{cases} \quad (6)$$

This prevents the educational algorithm from modelling all carbohydrates as if they entered the bloodstream immediately. In practical terms, it teaches the user why a high-fat meal may cause a late hyperglycaemic rise even when the early glucose response appears safe.

### 9.2 Sensor Lag and Derivative Control

Continuous Glucose Monitors (CGMs) measure glucose in interstitial fluid rather than directly in capillary blood. This creates a physiological delay, often discussed in the range of several minutes, between the true blood-glucose trend and the sensor-reported value [11]. To demonstrate this limitation, the IINTS MK2 PID controller gives explicit educational importance to the derivative term, which estimates the direction and speed of glucose change.

The simplified prediction logic can be expressed as:

$$BG_{\text{predicted}} = BG_{\text{CGM}} + \tau \cdot \frac{dBG}{dt}, \quad (7)$$

where  $\tau$  represents an assumed sensor-lag horizon. This allows the simulation to show that the system should not only ask “What is the glucose now?”, but also “Where is the glucose going?”

### 9.3 Circadian Rhythm and the Dawn Phenomenon

Insulin sensitivity fluctuates over a 24-hour cycle because of hormonal rhythms, including early-morning counterregulatory hormones such as cortisol. The resulting dawn phenomenon can increase insulin needs in the early morning [12]. To demonstrate this, the code includes a `get_circadian_isf()` function that dynamically lowers the insulin sensitivity factor (ISF) between 04:00 and 08:00:

$$ISF_{\text{effective}}(t) = \begin{cases} \alpha_{\text{dawn}} \cdot ISF_{\text{base}}, & 04:00 \leq t < 08:00, \\ ISF_{\text{base}}, & \text{otherwise,} \end{cases} \quad 0 < \alpha_{\text{dawn}} < 1. \quad (8)$$

Because correction insulin is inversely related to ISF, lowering the simulated ISF makes the model intervene more strongly against early-morning hyperglycaemia. This is presented as an educational demonstration of circadian physiology, not as a patient-specific dose rule.

### 9.4 Activity Mode and Physical Exercise

Physical activity can substantially increase insulin sensitivity and raise the risk of exercise-related hypoglycaemia in people with Type 1 Diabetes [13]. The IINTS MK2 therefore includes an `ACTIVITY_MODE`. When enabled, the algorithm immediately raises the safe target glucose value and reduces PID aggressiveness:

$$TARGET_{\text{activity}} = TARGET_{\text{base}} + \Delta_{\text{activity}}, \quad K_{p,i,d}^{\text{activity}} = \gamma \cdot K_{p,i,d}^{\text{base}}, \quad 0 < \gamma < 1. \quad (9)$$

This teaches that safe closed-loop control is context-dependent: during exercise, a glucose value that would normally justify correction may instead require caution, basal reduction, or a higher temporary target.

## 10 Qualitative Validation & Expected Outcomes

Rather than conducting a quantitative clinical test, which falls outside the scope of an educational prototype, the UX design of the IINTS MK2 was qualitatively validated through empirical observations. These insights are directly informed by the author’s personal field experience as a youth camp counsellor (“moni”) at camps for neurodivergent children, including children with Autism Spectrum Disorder (ASD) and ADHD.

During these camps, medical actions—such as operating a complex insulin pump—were observed to cause acute overstimulation, frustration, and loss of autonomy in some children. Based on this practical field experience, the expected outcomes of the IINTS “Tamagotchi” interface are as follows:

- **Reduction of cognitive overload:** By replacing abstract numbers with a familiar Bluey-inspired interface and photographs of meals, the task-initiation threshold for the child is expected to decrease substantially. A child may recognise “Pizza” faster than “60 g carbohydrates,” reducing uncertainty and performance anxiety.
- **Increased medical sovereignty:** In camp settings, children often hand over pump operation entirely to adults because they fear making mistakes with numerical inputs. The guided autonomous workflow allows the child to take the first step independently, strengthening ownership over their own body and care process.
- **Prevention of alarm fatigue:** Slow visual animations during simulated delivery replace shrill beeps, flashing warnings, and harsh alarm signals. This supports predictable and calm sensory processing, helping to prevent stress-related meltdowns at demanding moments such as mealtimes.

## 11 Cognitive Accessibility: Neurological and Visual Design

The most profound innovation of this educational prototype lies in the field of human–computer interaction (HCI) and psychology. Conventional medical devices are typically engineered by adults for clinical professionals, resulting in cold, number-heavy interfaces. For young children, and particularly neurodivergent users such as those with Autism Spectrum Disorder, ADHD, or dyslexia, this imposes an overwhelming cognitive load and sensory barrier [16, 17].

**Medical Sovereignty (Autonomy).** A critical flaw of standard insulin pumps is that their intimidating interfaces can render children entirely dependent on their parents to operate the machine. Abstract numerical data reduces the child to a passive “patient.” The primary goal of the IINTS interface is the restoration of medical sovereignty: by translating abstract medical data into the child’s own conceptual world, the child regains control and autonomy over their own body and disease management. They are no longer operating a “scary hospital machine,” but rather caring for a digital companion. This fosters independence, self-confidence, and disease acceptance at a significantly younger age.

**Bridging the Experiential Gap (The MK2 Paradigm).** Conventional insulin pumps are inherently foreign and intrusive to the experiential world of neurodivergent children and those with cognitive disabilities. To bridge this gap, the IINTS MK2 proposes integrating comforting, familiar cultural touchstones directly into the interface. For instance, the prototype explores the use of the widely recognised animated character “Bluey” as a digital companion. Because many children with varying degrees of cognitive disability find safety, regulation, and routine in familiar characters, adapting the medical device to their world—rather than forcing them to adapt to a clinical world—makes the pump significantly more manageable, approachable, and less alienating.



**Figure 2.** Example emotional glucose states used in the IINTS MK2 interface. The happy state represents a glucose value in range, the angry state represents hyperglycaemia, and the sad state represents hypoglycaemia.

**Dual Coding Theory and Metaphorical Learning.** The “Tamagotchi” mode actively reduces the dominance of intimidating clinical values, such as “120 mg/dL” or “30 g carbohydrates.” Instead, blood glucose levels are translated into facial expressions, such as happy, sleepy, or hyper, and carbohydrate inputs are represented by highly recognisable icons, such as an apple, a sandwich, or a pizza. Following Dual Coding Theory and Multimedia Learning principles [14, 15], coupling these visual metaphors with large, distinct text labels, such as “PIZZA,” reduces working-memory load. Even if the child is pre-literate or struggles with reading, the visual imagery supports word recognition and lowers the learning barrier.

**Low-Sensory and Neuro-Inclusive Design.** Neurodivergent children can be highly susceptible to visual and auditory sensory overload, which in a clinical context may contribute to alarm fatigue and severe stress responses. Reflecting universal-design principles [17], this interface deliberately avoids aggressive, overstimulating elements such as flashing progress bars, warning alarms, and harsh neon colours. Instead, it uses a calming palette of soft pastel colours. Furthermore, the typography uses high-contrast anthracite text instead of harsh pure black, softening the visual contrast and improving readability and comfort for dyslexic users.

## 12 Future Work

Because the IINTS pump is a purely educational model and not a certified medical device under medical-device regulation [21], future development focuses on educational and social impact rather than commercial clinical deployment.

The first priority is equal access to education. Patients, relatives, classmates, teachers, and healthcare students should be able to safely experiment with complex pump technology through a low-cost IINTS model. Many commercial diabetes technologies remain difficult to inspect, modify, or understand from an educational perspective. IINTS therefore explores how open educational models can make the underlying principles more accessible to students, patients, relatives, and healthcare learners.

The second priority is inclusion for neurodivergent users. This project acts as a call to action for the medical-technology industry: neurodivergent people must be considered from the beginning of medical interface design. Future work will therefore focus on broader design guidelines for low-sensory, cognitively accessible pump interfaces, so that accessibility becomes a default design requirement rather than an afterthought.

## 13 Conclusion

By combining actuator mechanics, simplified pharmacokinetic modelling, algorithmic reasoning, and low-sensory interface design, this project demonstrates how artificial pancreas technology can be transformed into an educational tool. The IINTS prototype does not claim to be a medical device; instead, it provides a safe way to learn, discuss, and visualise the scientific principles behind diabetes technology. In doing so, it argues that technical education, neurodivergent accessibility, and shared medical sovereignty should be central to the future of medical innovation.

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